Hybrid Search and Reranking

Improving retrieval relevance in document question-answering systems

Purpose

This short reference note explains retrieval methods that can improve a RAG pipeline when vector similarity alone does not return the best evidence. It is designed for semantic retrieval experiments and uses coherent, technical examples.

1. Why Semantic Search Sometimes Misses

Embedding-based retrieval can find passages with related meanings even when wording differs. However, semantic similarity is not the same as exact relevance. A question asking for an API name, version number, acronym, or error code may need an exact lexical match. Vector search can rank a conceptually related passage above the passage containing the precise term. This is why retrieval systems sometimes benefit from using more than one search method.

2. Lexical Search and BM25

Lexical search ranks documents using matching terms. BM25 is a widely used ranking method that considers term frequency, how rare a term is across the collection, and document length. It can work well for identifiers, technical vocabulary, and phrases that appear directly in a document. Its limitation is that synonyms and paraphrases may not match: a question using “speed up” may fail to match a passage that only says “reduce latency.” Lexical and semantic retrieval therefore have complementary strengths.

3. What Hybrid Search Means

Hybrid search combines lexical retrieval with vector-based semantic retrieval. One approach retrieves candidates from both methods and merges the results. A common fusion method is reciprocal rank fusion (RRF), which rewards documents that rank highly in multiple result lists without requiring raw scores to share the same scale. Another approach normalizes scores and combines them with chosen weights. The best method depends on the search engine, corpus, and query types; validate it on a labeled evaluation set.

4. Reranking Candidates

A reranker takes an initial list of candidate passages and scores how relevant each passage is to a specific query. A cross-encoder, for example, processes the query and passage together rather than embedding them independently. This can improve ordering but usually costs more computation than first-stage retrieval. A practical architecture retrieves a larger candidate set, reranks those candidates, and passes only the strongest few passages to the language model.

5. Example from Technical PDFs

Suppose a user asks, “Which setting controls the number of retrieved chunks?” Semantic search may return a general explanation of retrieval, while lexical search may find a passage containing “top-k.” Hybrid search can include both candidates. A reranker can then prioritize the passage that directly defines top-k. The generated answer should still be grounded in that passage and cite its source metadata. Retrieval and reranking improve candidate selection, but neither guarantees that the final answer is correct.

6. Practical Evaluation Plan

Create questions with different needs: paraphrases, exact identifiers, rare technical terms, and broad conceptual questions. For each question, label relevant source chunks. Compare vector-only retrieval, lexical-only retrieval, and hybrid retrieval using Recall@k or another suitable metric. Then measure whether reranking improves the ordering of relevant passages. Keep the same evaluation questions and corpus version across experiments. Also record latency and token usage, because a small relevance improvement may not justify a large operational cost.

7. Integration and Common Pitfalls

Start with a simple baseline and add hybrid retrieval only when error analysis shows a need. Ensure both search methods operate on the same current corpus and that duplicate candidates are merged correctly. Avoid combining raw scores blindly if they use different scales. Do not send every candidate to the language model: irrelevant passages increase prompt length and may distract generation. Preserve document IDs, chunk IDs, page numbers, and filenames throughout retrieval and reranking so citations remain accurate.

Key Takeaways

Vector search is strong at semantic similarity; lexical search is strong at exact terms. Hybrid search combines their candidate lists, while reranking improves the order of candidates using query–passage relevance. Evaluate each stage independently on representative questions before adding complexity to a document QA system.

End of document • Measure retrieval quality before changing production settings.